Ogbonna Arinze

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SUMMARY

Fullstack Engineer with 8 years of professional experience building production AI systems, real-time streaming infrastructure, and scalable TypeScript applications across React and Node.js. Proven track record architecting multi-agent LLM pipelines with RAG, vector search, and SSE streaming, alongside high-performance React frontends serving enterprise and consumer products. Passionate daily AI tool user with strong opinions on LLM orchestration, prompt engineering, and building AI-native software that delivers real value.

CORE COMPETENCIES

- **Fullstack TypeScript:** React, Next.js, Node.js, NestJS, Redux, React Query, Zustand, and Tailwind CSS.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, agentic workflows, and multi-agent orchestration.
- **Streaming & Real-time:** SSE streaming with keepalive and resumption, EventSource API, RxJS data streaming, and Redis-coordinated execution.
- **Databases & Infrastructure:** PostgreSQL (pgvector), MongoDB, Prisma, Docker, AWS, GitHub Actions, CI/CD pipelines, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright, Jest, Allure Reports, ESLint, and CI-driven end-to-end QA across multiple test projects.
- **Product Ownership:** End-to-end system design, architecture reviews, mentorship, performance optimization, and AI-native UI paradigms.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for connecting and transferring clinical trial sites' data to upstream EHR systems.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** with HTTP-only cookies for secure token handling, collaborating with backend engineers to align on auth strategies and enforce permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and optimizing the build pipeline for improved developer experience.
- Optimized application performance via **lazy loading, code splitting**, and **CDN asset delivery**, using **Lighthouse** and Chrome DevTools to identify and fix performance bottlenecks across devices.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, publishing results on GitHub Pages via **Allure Reports** and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

ActivEdge Technologies — Lagos, Nigeria
Senior React Developer — *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting transaction settlement rates by **80%** through innovative processing workflows.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Enhanced system performance by adopting the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Developed a real-time account reconciliation solution, boosting reconciliation of daily transactions by **60%** through innovative transaction processing and data management strategies.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **15+ REST API endpoints** for user management, payroll processing, and attendance tracking with comprehensive documentation.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with **5 specialized agents** (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) for three-stage intent classification (keyword scoring, LLM classification with 5-min cache, inferred fallback) and domain-specific RAG-powered analysis.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion (K=60), with **Cohere** rerank-2 reranking and **Voyage AI** voyage-3-large embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats every 15s to prevent reverse proxy timeouts, token-budgeted context management (summary + recent messages + user profile), conversation summarization, and memory extraction.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out Mastra-managed chunk tables from Prisma's drift detection.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 projects: web auth, dashboard auth, desktop Chrome, mobile Pixel 5), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and **Traefik** for reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*